Fabian Lander

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Education

PhD Mathematics *2024–May 2027 (expected)*
Max Planck Institute for Mathematics in the Sciences (IMPRS), degree from Leipzig University
Research Focus: Realizing the intrinsic geometry of surfaces, in two parts:
(i) Vertex-minimal paper tori (polyhedral embeddings of flat tori)
(ii) Surfaces of revolution in homogeneous three-manifolds
Advisor: JProf. Dr. James Farre

MSc Mathematics *2021–2023*
Heidelberg University, final grade 1.3 (German scale: 1.0 best, 4.0 lowest pass)
Thesis: Vertical Foliations of Triangulable Dilation Tori
Advisor: JProf. Dr. James Farre

BSc Mathematics *2017–2021*
Heidelberg University, final grade 1.8
Thesis: Polygonal Symplectic Billiards
Advisor: Prof. Dr. Peter Albers

Work Experience

Research Assistant (Mathematics Lab, Leipzig) *2024–Present*
Developing interactive visualizations for mathematical research and outreach
Collaborating with researchers on visual tools for geometry and dynamics
Supporting public engagement through exhibits and demonstrations

Research Assistant (University of Heidelberg/HEGL) *2022–2024*
Developed real-time visualization tools for flat surfaces using WebGL/Three.js
Supervised student projects in geometric computing
Collaborated on outreach events for HEGL Student Seminar

Research Assistant (University of Heidelberg/SFB 191) *2021–2024*
Developed high-performance symplectic billiard dynamics simulator (C++/OpenGL)
Co-authored a paper on periodic orbits in symplectic billiards
Implemented discrete dynamical system visualizations

Teaching Assistant in Analysis 1-3 (University of Heidelberg) *2020–2022*
Led tutorial sections of 20–30 students (two to four sessions per week) in
Real Analysis and Measure Theory, for lecture courses of about 300 students.

Publications and Preprints

A Universal 8-Vertex Origami Torus Triangulation, with Peter Doyle, Richard Evan Schwartz and Steve Trettel, preliminary version (2026), fabianlander.github.io/papers/universal-8-vertex.pdf.

The two most symmetric flat tori as eight-vertex paper tori, arXiv:2609.10703 (2026). 26 pages. Preprint.

Symplectic Tiling Billiards on Complete Affine Tori, with Charles Daly, arXiv:2608.28894 (2026). 47 pages. Preprint.

Illustrating Hyperbolic Surfaces with Mesh Embeddings, with Erik Löffelholz, Diaaeldin Taha, Steve Trettel and Anna Wienhard, in *Proceedings of Bridges 2026: Mathematics, Art, Music, Architecture, Culture*, pp. 197–204, Bridges Organization, 2026.

Immersive Visualization of Flat Surfaces Using Ray Marching, with Diaaeldin Taha, in *Proceedings of Bridges 2025: Mathematics and the Arts*, pp. 191–198, Bridges Organization, 2025.

Symplectic billiards for pairs of polygons, with Peter Albers and Jannik M. Westermann, arXiv:2402.12244

(2024). 38 pages. Accepted for publication in the *Arnold Mathematical Journal*.

Illustration als mathematische Forschungstechnik, with Tim Reinhardt and Martin Skrodzki, *Mitteilungen der Deutschen Mathematiker-Vereinigung* 34 (2026), no. 3, 153–158.

Selected Software Projects

Eight-Vertex Paper Tori Verification *Python, 2026*
Exact-arithmetic verification of every claim in the eight-vertex paper tori preprint, over $\mathbb{Q}$ and $\mathbb{Q}(\sqrt{3})$
github.com/FabianLander/eight-vertex-paper-tori-verification

Interactive Pseudosphere with Hyperbolic Tiling *Three.js/GLSL, 2026*
Real-time (2,3,7) triangle tiling on surfaces of revolution with adjustable ambient curvature, using RK4 ODE integration
fabianlander.github.io/talks/rotationally-symmetric-surfaces

Flat Tori Gallery *Three.js/WebGL, 2026*
Interactive isometric embedding experiments of flat tori in $\mathbb{R}^3$ and S^3
fabianlander.github.io/demos

Symplectic Billiards Simulator *C++/OpenGL, 2020*
High-performance research tool for dynamical systems analysis
github.com/SiegfriedFabian/Billiards